

1. Domain Name System (DNS) Architecture

Port 53 (UDP/TCP)

- **Core Purpose:** Distributed, hierarchical database translating human-friendly domain names (e.g., `aktu.ac.in`) into 32-bit/128-bit IP addresses.
- **Transport Ports:** UDP Port 53 for interactive standard queries ($\leq 512B$); TCP Port 53 for zone transfers and extended replies ($> 512B$).
- **3 Domain Space Divisions:**
 - *Generic Domains (gTLD):* `.com` (commercial), `.edu` (education), `.gov` (govt), `.org` (non-profit), `.net`.
 - *Country Domains (ccTLD):* 2-letter country code: `.in`, `.us`, `.uk`.
 - *Inverse Domains (PTR):* Reverse mapping (IP $\rightarrow$ Hostname) via `in-addr.arpa`.
- **Recursive vs Iterative Resolution:**
 - *Recursive:* Queried server takes full burden to resolve and return final IP to client stub resolver.
 - *Iterative:* Queried server returns best-effort referral IP pointer to next subordinate nameserver.
- **Caching & TTL:** Answers cached locally for `TTL` seconds to reduce internet traversal delay.

2. DNS Resource Records (RR) Anatomy

Format

<Domain_Name, Value, Type, TTL>			
Type	Mapping Target	Practical Example	
A	Hostname $\rightarrow$ IPv4 (32-bit)	<code>aktu.ac.in $\rightarrow$ 14.139.233.10</code>	
AAAA	Hostname $\rightarrow$ IPv6 (128-bit)	<code>aktu.ac.in $\rightarrow$ 2404:6800::</code>	
CNAME	Alias $\rightarrow$ Canonical Name	<code>www.aktu.ac.in $\rightarrow$ aktu.ac.in</code>	
MX	Domain $\rightarrow$ Mail Exchanger	<code>aktu.ac.in $\rightarrow$ mail.aktu.ac.in</code>	
NS	Zone $\rightarrow$ Authoritative Server	<code>aktu.ac.in $\rightarrow$ ns1.aktu.ac.in</code>	
PTR	IP $\rightarrow$ Hostname (Reverse)	<code>10.233.139.14.in-addr.arpa</code>	

3. World Wide Web & HTTP / HTTPS

Port 80 vs 443

- **URL Syntax:** `protocol://host:port/path` (e.g., `http://aktu.ac.in:80/syllabus`).
- **Stateless Nature:** HTTP maintains zero client history between requests. Stateful sessions managed via **Cookies** and **JWT**.
- **HTTP Methods:** `GET` (fetch resource, idempotent), `POST` (submit payload/data), `HEAD` (headers only), `PUT` (create/replace), `DELETE`.
- **Status Families:** 1xx (Info), 2xx (Success: `200 OK`), 3xx (Redirection: `301 Moved`, `304 Not Modified`), 4xx (Client Error: `400 Bad`, `404 Not Found`), 5xx (Server: `500 Error`).
- **Persistent vs Non-Persistent HTTP:**
 - *HTTP/1.0 (Non-Persistent):* 1 TCP connection per object (2 RTT overhead).
 - *HTTP/1.1 (Persistent):* Single TCP connection for all objects (Connection: `keep-alive`) + Pipelining.
- **HTTP vs HTTPS:** HTTPS runs HTTP over **TLS/SSL (Port 443)**; provides AES encryption, HMAC integrity, and CA Certificate authentication.

4. Remote Login & TELNET Protocol

Port 23 (RFC 854)

- **Local vs Remote Login:** Local login interacts directly with local terminal driver; Remote login sends keystrokes over TCP to remote pseudoterminal driver (PTY).
- **Network Virtual Terminal (NVT):** Universal intermediate representation. 7-bit ASCII characters (0–127); 8th bit high for control commands.
- **Commands:** Starts with **IAC (Interpret As Command = 255 / 0xFF)**. `WILL`, `DO`, `WONT`, `DONT` for option negotiation.
- **Modes:** Default (character echo), Character (keystroke per packet), Line (local buffer edited before Enter).
- **Vulnerability:** Cleartext passwords! Completely replaced by **SSH (Port 22)**.

5. Electronic Mail (E-Mail) Architecture & MIME

4 Tiers

- **4 Scenarios:** 1 (Same system), 2 (Direct connected hosts), 3 (Alice remote, Bob local), 4 (Modern: Both connected to independent mail servers).
- **Triad Components:** User Agent (UA - Compose/Read), Message Transfer Agent (MTA - SMTP Push), Message Access Agent (MAA - POP3/IMAP Pull).
- **MIME (RFC 2045):** Overcomes SMTP 7-bit ASCII limitation; encodes multimedia/binary into ASCII. Headers: `MIME-Version`, `Content-Type`, `Content-Transfer-Encoding` (Base64 ratio 3 raw bytes $\rightarrow$ 4 ASCII chars).

6. Simple Mail Transfer Protocol (SMTP)

Port 25 / 587

- Protocol Nature:** Strictly a **PUSH Protocol** executing over TCP Port 25 (server relay) and Port 587 (client submission).
- 3-Phase Operational Cycle:**
 - 1. *Connection:* Server sends 220 → Client sends HELO / EHLO → Server sends 250 OK .
 - 2. *Mail Transaction:*
 - MAIL FROM: <sender@domain> → 250 OK
 - RCPT TO: <recv@domain> → 250 OK
 - DATA → 354 Start mail input → Body text → <CRLF>.<CRLF> → 250 Queued .
 - 3. *Teardown:* Client sends QUIT → Server sends 221 Closing .
- Relays & MX Lookup:** Messages route through intermediate MTAs using DNS MX priority preference records.
- Email Security Gateways (AKTU PYQ):** On-premise appliances (Cisco IronPort, Barracuda, Fortinet FortiMail, Symantec) and Cloud (Proofpoint, Mimecast) for spam filtering, DLP, and antivirus scanning.

7. Mail Access Protocols: POP3 vs IMAP4

PULL Engine

Parameter	POP3 (RFC 1939)	IMAP4 (RFC 3501)
Ports	Port 110 (995 SSL)	Port 143 (993 SSL)
Storage	Local client disk	Central server database
Sync	No multi-device sync	Full real-time synchronization
Modes	Delete Mode & Keep Mode	Continuous server stateful
Flags	Local flags only	\Seen , \Flagged synced
Folders	No server folders	Create/manage server folders
Partial Fetch	Full message download	Header preview & search

8. File Transfer Protocol (FTP RFC 959)

Dual Connections

- Why Dual Connections?** Separation of concerns: control uses 7-bit ASCII lines; data transfers raw bytes without command choking. Allows ABOR mid-transfer.
- Control Connection (Port 21):** Stays open throughout session. 7-bit ASCII commands (USER , PASS , RETR , STOR , QUIT) and 3-digit replies.
- Data Connection (Port 20 / PASV):** Opens dynamically per file transfer / directory list; closes immediately upon completion.
- Active Mode (PORT):** Client sends PORT IP ,p1,p2 . Server initiates data connection FROM Port 20 TO Client. *Fails behind client NAT firewalls!*
- Passive Mode (PASV):** Client sends PASV . Server opens random high port \$P\$ (\$>1023\$) → Client connects TO Server. *Firewall friendly!*
- Port Math:** $\$ \text{Port} = p_1 \times 256 + p_2$. (e.g., \$(16, 32)\$ implies $16 \times 256 + 32 = 4128$).
- Anonymous FTP:** Username anonymous , Password = user email.

9. TFTP vs FTP vs HTTP Comparison

RFC 1350

- TFTP (Trivial FTP):** Lightweight, UDP Port 69. No authentication. Fixed 512-byte blocks with Stop-and-Wait ACK. Block size < 512B signals EOF. Used in PXE network boot and router IOS firmware flashing.

Feature	FTP (RFC 959)	TFTP (RFC 1350)	HTTP (RFC 2616)
Transport	TCP 20 & 21	UDP 69	TCP 80 / 443
Channels	Dual Connections	Connectionless	Single Stream
Auth	Mandatory	None (Zero)	Header / Token
Directory	Full browsing	No listing	URL paths
Footprint	Heavy daemon	Fits in ROM/BIOS	Rich client/cache

10. Network Management & SNMP

UDP 161 & 162

- **Framework:** Manager (client on NMS) polls Agents (servers on routers/switches).
- **Ports:** UDP 161 (Agent receives Get/Set); UDP 162 (Manager receives Traps).
- **Management Triad (Slide 65 Analogy):**
 - *SMI:* Grammar/syntax rules using ASN.1 and BER encoding.
 - *MIB:* Object database tree. Prefix: 1.3.6.1.2.1 (mib-2).
 - *SNMP:* Protocol packet formats and message passing.
- **3 Functions:** Status Polling (Get), Remote Configuration (Set), Emergency Alarms (Trap).

11. Cryptography Foundations & Symmetric Ciphers

DES & AES

- **4 Pillars:** Confidentiality, Integrity, Authentication, Non-repudiation.
- **Symmetric:** Single shared key \$K\$. \$C = E_K(P)\$, \$P = D_K(C)\$. Key distribution challenge: \$N(N-1)/2\$ keys.
- **Classical:** Substitution (Caesar shift \$k=3\$, Monoalphabetic vs Polyalphabetic) vs Transposition (Rail Fence, Columnar).
- **DES:** 64-bit block, 56-bit key, 16 Feistel rounds (Broken).
- **AES:** 128-bit block, 128/192/256-bit key, SPN network (SubBytes, ShiftRows, MixColumns, AddRoundKey). Global gold standard.

12. Data Compression Techniques

Lossless & Lossy

- **Compression Ratio:** $\text{CR} = \frac{\text{Uncompressed}}{\text{Compressed}}$.
- **Lossless (Exact bits):**
 - *Run-Length (RLE):* AAAAABBBCCDA → A5B3C2D1A1 .
 - *Huffman:* Variable-length prefix codes based on character probabilities.
 - *Lempel-Ziv (LZW):* Dynamic dictionary table index pointers (ZIP/PNG).
- **Lossy Audio Masking (Slide 87–88):**
 - *Frequency Masking:* Louder tone drowns nearby soft frequencies.
 - *Temporal Masking:* Loud impulse (hammer) temporarily desensitizes ear.

13. RSA Asymmetric Algorithm & Math

AKTU PYQ

- **Key Selection Steps:**
 1. Choose distinct large primes \$p\$ and \$q\$.
 2. Modulus: \$n = p \times q\$.
 3. Totient: \$\phi(n) = (p - 1)(q - 1)\$.
 4. Public exponent \$e\$: \$1 < e < \phi(n)\$ and \$\gcd(e, \phi(n)) = 1\$.
 5. Private exponent \$d\$: \$d \cdot e \equiv 1 \pmod{\phi(n)} \implies d = e^{-1} \pmod{\phi(n)}\$.
 6. Public Key = \$(e, n)\$, Private Key = \$(d, n)\$.
- **Formulas:** \$C = P^e \bmod n\$ and \$P = C^d \bmod n\$ (where \$P < n\$).
- **Solved Example (Character "F"):**
 - \$p=3, q=11 \implies n=33, \phi(n)=20\$.
 - Choose \$e=3 \implies \gcd(3, 20)=1\$.
 - \$3d \equiv 1 \pmod{20} \implies 3 \times 7 = 21 \implies d=7\$.
 - Plaintext \$P = 5\$ ('F').
 - **Encrypt:** \$C = 5^3 \bmod 33 = 125 \bmod 33 = 26\$.
 - **Decrypt:** \$P = 26^7 \bmod 33 = 5\$ ('F') ✓.

14. Firewalls & 5 Diagnostic Commands

Slides 89–93

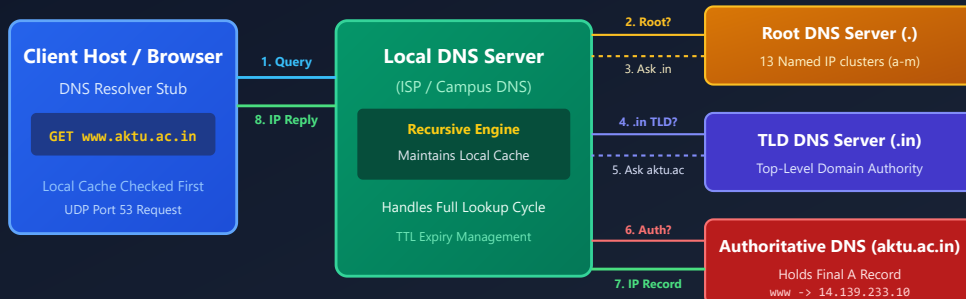
- **Firewalls:** Packet Filtering (L3/L4 stateless IP/Port matching), Stateful Inspection (L4 connection state table), Application Proxy (L7 payload inspection).
- **Digital Signatures:** Hash signed with sender's Private Key \$S = E_{\text{PR}_A}(H(M))\$; verified with Public Key.
- **5 Networking Commands:**
 - `ipconfig /all` : Shows IP, MAC, Subnet, Gateway, DNS; `/flushdns` .
 - `nslookup <domain>` : Resolves hostname to IP & queries MX/A records.
 - `tracert <target>` : Incremental IP TTL hops & latency discovery.
 - `ping <target>` : ICMP Echo Request (Type 8) & Reply (Type 0).
 - `netstat -ano` : Displays active TCP links, listening ports & PIDs.

Module 01: Domain Name System (DNS) Architecture

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

DOMAIN NAME SYSTEM (DNS) RESOLUTION ARCHITECTURE

Hierarchical Tree, Recursive vs Iterative Resolution, Resource Records & Caching



DNS PROTOCOL SPECIFICATIONS & RESOURCE RECORD (RR) ARCHITECTURE

1. Domain Tree Hierarchy

- Root Domain: Denoted by dot (.)
- TLD: Generic (.com, .edu, .org, .gov)
- ccTLD: Country Code (.in, .us, .uk, .jp)
- Second-Level / Subdomains: aktu.ac.in

2. Resolution Types

- Recursive Query: Server does full lookup and returns final answer to client.
- Iterative Query: Server returns referral IP pointer to next lower server.

3. Key Resource Records (RR)

- A: Hostname -> IPv4 (32-bit)
- AAAA: Hostname -> IPv6 (128-bit)
- CNAME: Alias canonical domain name
- MX: Mail exchange server | NS: Nameserver

1. Domain Name System (DNS) Overview

DNS translates human-readable hostnames into machine-routable IP addresses using a distributed, hierarchical database.

Protocol Standards & Port Assignment

Transport Protocol: UDP Port 53 for standard interactive queries (≤ 512 bytes). TCP Port 53 for zone transfers and extended EDNS0 responses (> 512 bytes).

2. Domain Name Space Divisions

Division	Scope & Syntax	Standard Examples
Generic Domains (gTLD)	Categorizes registered institutions by organizational nature	.com (Commercial), .edu (Educational), .org (Non-profit), .gov (Government)
Country Domains (ccTLD)	Two-letter geographic national country codes	.in (India), .us (United States), .uk (United Kingdom), .jp (Japan)
Inverse Domains (PTR)	Reverse mapping: Translates known IP address back into hostname	in-addr.arpa special domain zone for IPv4 reverse resolution

3. Recursive vs Iterative Query Resolution

Criterion	Recursive Resolution	Iterative Resolution
Client Expectation	Demands definitive final answer or explicit failure error	Accepts best-effort referral pointer to subordinate server
Processing Workload	Heavy computation borne by the interrogated server	Distributed across querying client / recursive ISP resolver
Topology Placement	Client Stub Resolver $\leftrightarrow$ Local ISP DNS Server	Local ISP DNS Server $\leftrightarrow$ Root $\rightarrow$ TLD $\rightarrow$ Authoritative Servers

4. Core Resource Record (RR) Types

Format: <Domain_Name, Value, Type, TTL>

- **A Record:** Resolves hostname to 32-bit IPv4 address.

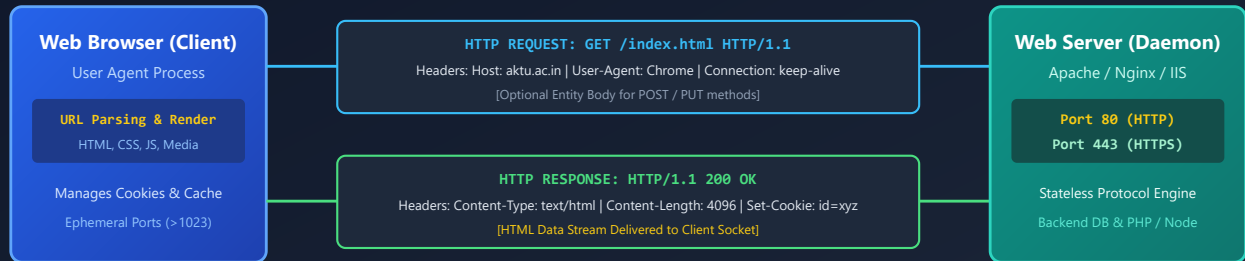
- **AAAA Record:** Resolves hostname to 128-bit IPv6 address.
- **CNAME Record:** Canonical alias pointing alias domain to master domain.
- **MX Record:** Identifies primary and backup mail exchange routing servers.
- **NS Record:** Delegates DNS zone authority to specified nameservers.

Module 02: World Wide Web & HTTP / HTTPS Protocols

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

WORLD WIDE WEB: HTTP & HTTPS TRANSACTION ARCHITECTURE

Request-Response Cycle, Methods, Status Codes, Persistent vs Non-Persistent & SSL/TLS



1. Methods & Status Codes

GET: Fetch resource (idempotent) • **POST:** Submit data / upload payload • **PUT:** Replace resource | **DELETE:** Remove • **200 OK** | **301 Moved** | **304 Not Modified** • **400 Bad Request** | **404 Not Found** • **500 Server Error** | **503 Unavailable**

2. Persistent vs Non-Persistent • **HTTP/1.0 (Non-Persistent):** 1 TCP connection per object (2 RTT delay) • **HTTP/1.1 (Persistent):** Single TCP connection for all webpage objects • **Pipelining:** Send next request without waiting for previous object's ACK

3. HTTP vs HTTPS Security • **HTTP (Port 80):** Plaintext data transfer. Vulnerable to sniffing and MITM attacks. • **HTTPS (Port 443):** HTTP over TLS/SSL. Asymmetric handshake + Symmetric encryption. • **Digital Certificate:** Authenticates server issued by trusted Certificate Authority (CA).

1. HTTP Fundamentals & Stateless Architecture

Hypertext Transfer Protocol is the foundation of data communication for the World Wide Web, operating as a request-response protocol over TCP.

Stateless Character of HTTP

HTTP does not preserve client conversational state between successive requests. To maintain stateful interactions (shopping carts, user logins), modern web architectures use **HTTP Cookies** and **JWT Tokens**.

2. HTTP Methods & Status Code Families

Code Range	Category	Representative Codes
1xx	Informational	100 Continue , 101 Switching Protocols
2xx	Success	200 OK , 201 Created , 204 No Content
3xx	Redirection	301 Moved Permanently , 304 Not Modified (Cache validation)
4xx	Client Errors	400 Bad Request , 401 Unauthorized , 403 Forbidden , 404 Not Found
5xx	Server Errors	500 Internal Server Error , 502 Bad Gateway , 503 Service Unavailable

3. HTTP vs HTTPS Comparison

Parameter	HTTP	HTTPS
Default Port	Port 80	Port 443

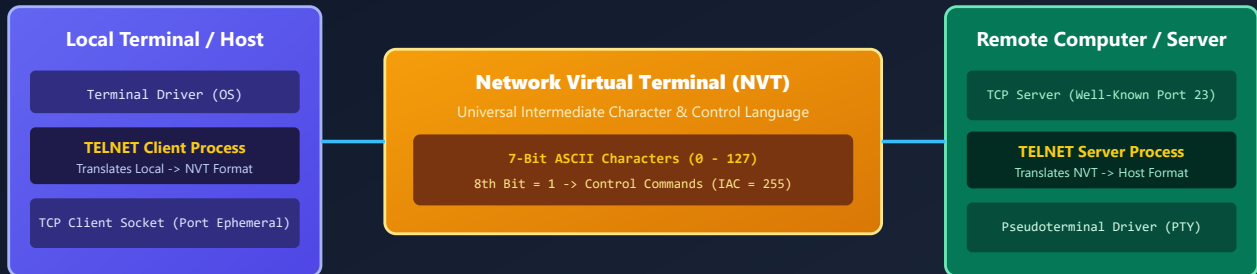
Security Transport	Plaintext TCP socket stream	Encrypted over TLS/SSL handshake tunnel
Confidentiality & Integrity	Vulnerable to eavesdropping & MITM attacks	AES symmetric encryption + HMAC integrity verification

Module 03: Remote Login & TELNET Architecture

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

TELNET PROTOCOL & REMOTE LOGIN ARCHITECTURE

Local vs Remote Login, Network Virtual Terminal (NVT), Option Negotiation & SSH Comparison



1. Local vs Remote Login

Local Login: Keystrokes handled by terminal driver directly to local OS shell. • **Remote Login:** Local terminal behaves as terminal for remote time-sharing host. • Transparent keystroke transmission over TCP. 2. NVT & Operating Modes • **Default Mode:** Half-duplex character-at-a-time transmission with remote echoing. • **Character Mode:** 1 char sent per packet. • **Line Mode:** Lines edited locally and transmitted in bulk on pressing ENTER. 3. TELNET vs SSH Security • **TELNET (Port 23):** Cleartext transmission! Passwords & root commands visible to packet sniffers. • **SSH - Secure Shell (Port 22):** Encrypted session tunnel using public keys and AES. Replaced TELNET globally.

1. Remote Login Paradigm

Remote login allows a terminal user on a local system to log in to and execute commands on a remote timesharing system across an IP internetwork.

The Role of Network Virtual Terminal (NVT)

To overcome terminal and keyboard heterogeneity across operating systems, TELNET maps local terminal characteristics to a universal intermediate representation called **NVT**.

2. TELNET Control Commands & Option Negotiation

Command	Byte Code	Semantic Action
IAC	255 (0xFF)	Interpret As Command (signals that subsequent byte is a TELNET control token)
WILL	251	Sender offers to initiate or enable a specific terminal capability
DO	253	Receiver requests sender to activate a specific terminal feature
WONT / DONT	252 / 254	Negative negotiation (refusal to enable requested feature)

3. TELNET vs SSH Security Analysis

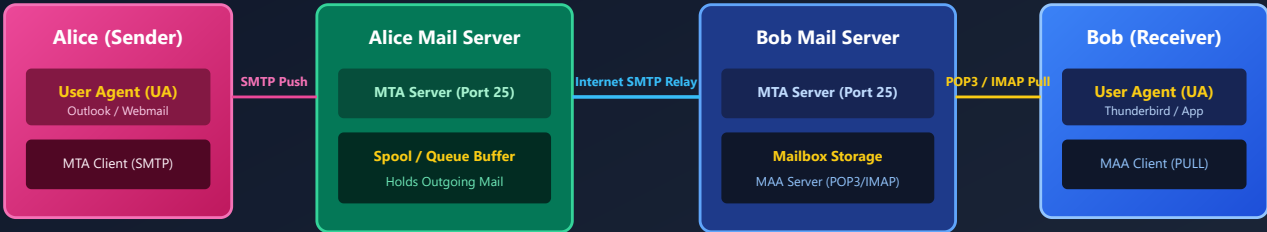
Property	TELNET (Port 23)	SSH (Port 22)
Transport Security	Plaintext stream without authentication integrity	Encrypted over cryptographic tunnel with host verification
Threat Exposure	Vulnerable to sniffing and credentials interception	Immune to passive wiretapping and active session hijacking

Module 04: Electronic Mail Architecture and MIME

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

ELECTRONIC MAIL (E-MAIL) 4-TIER ARCHITECTURE & MIME

User Agent (UA), Message Transfer Agent (MTA / SMTP), Message Access Agent (MAA / POP3 / IMAP) & MIME



1. Four E-Mail Scenarios

Scenario 1: Both on same timesharing host. • **Scenario 2:** Direct connection between 2 hosts. • **Scenario 3:** Alice remote, Bob local. • **Scenario 4 (Modern Web):** Both Alice & Bob connected to mail servers. Requires SMTP for PUSH + POP3/IMAP for PULL.

2. Core Architectural Roles • **User Agent (UA):** Client UI to compose, read, and organize email messages. • **Message Transfer Agent (MTA):** SMTP daemon pushing mail from sender to destination server. • **Message Access Agent (MAA):** POP3 / IMAP pulling mail from mailbox to recipient machine. 3. MIME (RFC 2045 - 2049) • **SMTP Limitation:** Can only send 7-bit ASCII. • **MIME Solution:** Encodes binary, audio, video into 7-bit ASCII before passing to SMTP. • **MIME-Version:** 1.0 • **Content-Type:** text, image, audio, multipart • **Transfer-Encoding:** Base64, Quoted-Printable

1. Electronic Mail 4-Scenario Evolutionary Framework

E-mail relies on a store-and-forward architecture transitioning from local single-system file append to a 4-tier distributed internet model.

The Fundamental Architectural Rule

SMTP is exclusively a **PUSH protocol** (sending client → server → server). Retrieving stored messages from a mailbox requires a **PULL protocol** (POP3 or IMAP4).

2. Functional Component Triad

Component	Abbr.	Role & Protocol Assignment
User Agent	UA	Client application interface for composing, reading, and managing mail (CLI: mailx, GUI: Outlook/Webmail).
Message Transfer Agent	MTA	Daemon responsible for relaying mail across the IP network using SMTP (Port 25) .
Message Access Agent	MAA	Client-server retrieval mechanism for pulling stored mail to client using POP3 (Port 110) or IMAP4 (Port 143) .

3. MIME Header Fields (RFC 2045)

Base64 Encoding Ratio: 3 raw bytes (24 bits) → 4 printable ASCII characters (33% size expansion)

- **MIME-Version:** Identifies protocol iteration (Standard value: 1.0).
- **Content-Type:** Declares media type and subtype (e.g., `multipart/mixed` , `image/jpeg`).

- **Content-Transfer-Encoding:** Declares transformation scheme (`base64` , `quoted-printable`).

Module 05: Simple Mail Transfer Protocol (SMTP) and Gateways

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

SIMPLE MAIL TRANSFER PROTOCOL (SMTP) & GATEWAYS

Connection Establishment, Mail Transaction, Teardown & Enterprise Security Gateways

Sender Mail Server

MTA Client Engine

Outbox Spool

Queued Outgoing Mail

Pushes Messages

Port 25 / 587 (Submission)

SMTP TRANSACTION 3-PHASE INTERACTION (TCP PORT 25)

```
<-- 220 server.com Simple Mail Transfer Ready
--> HELO client.domain.com (or EHLO)
<-- 250 Hello client.domain.com, pleased to meet you
--> MAIL FROM: <alice@aktu.ac.in>
<-- 250 2.1.0 Sender OK
--> RCPT TO: <bob@aktu.ac.in> | <-- 250 Recipient OK
--> DATA | <-- 354 Start mail input; end with <CRLF>.<CRLF>
[Body text / MIME payload...] | --> . | <-- 250 OK: queued
--> QUIT | <-- 221 2.0.0 server.com Service closing transmission
```

Receiver Mail Server

MTA Server Daemon

User Mailbox

Stores Delivered Mail

Listening on Port 25

Enforces Spam Checks

1. Key SMTP Commands

HELO / EHLO: Client identifies identity • **MAIL FROM:** Specifies envelope sender • **RCPT TO:** Specifies destination mailbox • **DATA:** Initiates message payload transfer • **QUIT:** Gracefully tears down TCP link

2. Mail Relays & DNS MX • **Mail Relays:** Intermediate MTAs forwarding emails across autonomous corporate networks. • **DNS MX Lookup:** Resolves destination domain to highest priority mail host IP. • Multiple MX records provide auto-failover.

3. Email Gateways (AKTU PYQ) • **Purpose:** Filtering, virus scan & routing. • **On-Premises / Appliance Gateways:** Cisco IronPort, Barracuda Spam Gateway, Fortinet FortiMail, Symantec Mail Security. • **Cloud Gateways:** Proofpoint, Mimecast.

1. SMTP Operational Paradigm

SMTP is a reliable, push-based application protocol executing over TCP Port 25 for inter-MTA communication and Port 587 for client submission.

Three Discrete Interactive Phases

Every SMTP transmission sequence proceeds through: **1. Connection Establishment** (HELO/EHLO), **2. Mail Transaction** (MAIL FROM, RCPT TO, DATA), and **3. Teardown** (QUIT).

2. Primary SMTP Commands & Status Replies

Command	Server Reply	Semantic Meaning
HELO client.domain	250 OK	Identifies client host identity to the server
MAIL FROM: <sender>	250 Sender OK	Specifies message envelope origin
RCPT TO: <recipient>	250 Recipient OK	Declares destination mailbox destination
DATA	354 Start mail input	Initiates raw text/MIME message body streaming
QUIT	221 Service closing	Graceful session termination

3. Email Gateways (AKTU PYQ Classification)

- **Appliance / On-Premises:** Cisco IronPort (ESA), Barracuda Email Security, Fortinet FortiMail, Symantec Messaging Gateway.
- **Cloud-Hosted:** Proofpoint Enterprise Protection, Mimecast Cloud Gateway.

Module 06: Mail Access Protocols: POP3 vs IMAP4

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MESSAGE ACCESS PROTOCOLS: POP3 vs IMAP4 ARCHITECTURE

Mail Retrieval Mechanics, Delete vs Keep Mode, Server-Side Folder Synchronization & Bandwidth

POP3 (Post Office Protocol v3 — RFC 1939)

Port: TCP Port 110 (Cleartext) | Port 995 (POP3S over SSL) • **Core Philosophy:** Simple offline download-and-read model. • **Delete Mode:** Mailbox mail deleted from server upon download. • **Keep Mode:** Copies kept on server, but states not synchronized. • **Single Device Bound:** Multi-device usage causes state divergence. • **No Folder Hierarchy:** Cannot create or organize server-side folders. • **Full Download Only:** Must download full message including attachments. Commands: USER, PASS, STAT, LIST, RETR, DELE, QUIT IMAP4 (Internet Message Access Protocol v4 — RFC 3501) • **Port:** TCP Port 143 (Cleartext) | Port 993 (IMAPS over SSL) • **Core Philosophy:** Dynamic client-server synchronized database. • **Multi-Device Sync:** Phone, laptop & webmail see identical mailbox state. • **Message Flags:** \Seen, \Answered, \Flagged, \Deleted tracked centrally. • **Server Folders:** Create, rename, delete hierarchical folders on server. • **Partial Fetch:** Download headers or preview without 20MB attachments. • **Server-Side Search:** Search keywords without downloading all emails. Commands: LOGIN, SELECT, FETCH, STORE, COPY, SEARCH, LOGOUT MASTER COMPARATIVE MATRIX: POP3 vs IMAP4 1. Protocol Role & Ports • **Type:** Both are Message Access (PULL) • **POP3:** Port 110 (Clear) / 995 (SSL) • **IMAP4:** Port 143 (Clear) / 993 (SSL) • **SMTP Role:** Pushes mail; cannot pull. 2. Storage & Sync Mechanics • **POP3:** Local client machine storage. • **IMAP4:** Central mail server storage. • **Read Status:** POP3 cannot sync read tags; IMAP4 syncs across all logged-in devices. 3. Bandwidth & Resource Efficiency • **POP3:** Requires full file download. • **IMAP4:** Headers-only preview saves data. • **Server Load:** POP3 has very low RAM churn; IMAP4 requires robust high-RAM servers.

1. Message Access Paradigm (PULL vs PUSH)

While SMTP handles inter-server PUSH operations, client devices require PULL protocols (POP3 or IMAP4) to retrieve messages from intermediate mailbox buffers.

Architectural Divergence

POP3 views the mail server as a temporary post office drop box; IMAP4 views the server as a perpetual, synchronized cloud database.

2. POP3 Operational Modes

- **Delete Mode:** Mail is transferred to client hard drive and immediately purged from server spool.
- **Keep Mode:** Messages persist on the server following retrieval, but message state changes remain strictly local.

3. Master Comparative Breakdown: POP3 vs IMAP4

Criterion	POP3 (RFC 1939)	IMAP4 (RFC 3501)
Well-Known Ports	Port 110 (Plain) / 995 (SSL)	Port 143 (Plain) / 993 (SSL)
Primary Storage Target	Client host disk	Server-side database storage

Multi-Device Consistency	Fails across multiple concurrent devices	Maintains real-time status synchronization
Partial Retrieval	Requires complete segment download	Allows isolated header and MIME part preview
Server Complexity	Minimal memory and disk utilization	Resource intensive (CPU, indexing, disk quota)

Module 07: File Transfer Protocol (FTP) Architecture

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

FILE TRANSFER PROTOCOL (FTP) DUAL-CONNECTION ARCHITECTURE

Separation of Control (Port 21) & Data (Port 20), Active Mode (PORT) vs Passive Mode (PASV)



Separation of Concerns: Control uses simple 7-bit ASCII lines; Data transfers raw bytes. • **Interactive Control:** User can abort a huge 10GB file transfer mid-way via Port 21. • Prevents command choking during file transfer. 2. Active FTP Mode (PORT) • Client connects to Server Port 21. • Client sends PORT IP,p1,p2 command. • **Server initiates data connection FROM Port 20 TO Client's Port N+1.** • **Firewall Problem:** Client NAT blocks server's incoming SYN! 3. Passive FTP Mode (PASV) • Client sends PASV command to Server. • Server opens random port P (>1023) and replies: 227 Entering Passive Mode (IP,p1,p2) . • **Client initiates data connection TO Server.** • **Firewall Friendly:** Outbound connections allowed!

1. FTP Dual-Connection Architecture

File Transfer Protocol establishes two discrete logical connections over TCP: an interactive Control Connection and an ephemeral Data Connection.

Functional Connection Roles

Control Connection (Port 21): Stays persistently open throughout session; transmits 7-bit ASCII commands and 3-digit replies.

Data Connection (Port 20 / PASV): Opens dynamically per file transfer / directory listing; closes automatically upon completion.

2. Active Mode vs Passive Mode FTP

Criterion	Active Mode (PORT)	Passive Mode (PASV)
Data Connection Initiator	Server initiates connection to Client	Client initiates connection to Server
Server Data Port	Fixed Well-Known Port 20	Ephemeral dynamic unreserved port (>1023)
Client Firewall Traversal	Fails: Client NAT firewall blocks unsolicited server SYN	Succeeds: Outbound connection permitted by firewall state tables

3. Key FTP Commands

- USER / PASS :** Client authentication credentials.
- PORT / PASV :** Selects active vs passive data connection negotiation mode.
- RETR / STOR :** Retrieves file from server / stores local file to server.
- QUIT :** Gracefully closes control connection.

Module 08: Trivial File Transfer Protocol (TFTP) vs FTP and HTTP

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

TRIVIAL FTP (TFTP) vs FTP vs HTTP ARCHITECTURAL MATRIX

UDP Bootstrapping Mechanics, 512-Byte Blocks, Stop-and-Wait ACK & Protocol Comparison

TFTP Architecture (RFC 1350 — UDP Port 69)

Transport Protocol: UDP Port 69 (Minimalist, connectionless). • **No Authentication:** Zero username / password security check. • **Fixed 512-Byte Data Blocks:** Data split into fixed 512-byte payload. • **Stop-and-Wait Error Control:** Sender sends Block N, waits for ACK N. • **Termination Signal:** Block size < 512 bytes signals EOF (End of File). • **Primary Use-Case:** PXE Network Booting, loading Cisco IOS firmware. 5 Packet Types: RRQ (1), WRQ (2), DATA (3), ACK (4), ERROR (5) TFTP STOP-AND-WAIT INTERACTION Client (TID) Server (69) RRQ (Read Request "firmware.bin") DATA Block 1 (512 Bytes) ACK Block 1 DATA Block 2 (214 Bytes < 512 = EOF) 3-WAY MASTER ARCHITECTURAL COMPARISON: FTP vs TFTP vs HTTP FTP (RFC 959) • **Transport:** TCP Ports 20 & 21 • **Connections:** Dual (Control & Data) • **Security:** User/Pass auth support • **Directories:** Full browsing, rename, delete • **Memory Footprint:** Heavy implementation TFTP (RFC 1350) • **Transport:** UDP Port 69 • **Connections:** Connectionless datagrams • **Security:** None (No auth / cleartext) • **Directories:** Cannot list or change dirs • **Footprint:** Ultra-light (Fits in ROM/BIOS) HTTP (RFC 2616) • **Transport:** TCP Port 80 / 443 • **Connections:** Single TCP stream per session • **Security:** HTTPS SSL/TLS encryption • **Primary Use:** Hypertext Web & REST APIs • **Footprint:** Rich feature stack & caching

1. Trivial FTP Architecture & Minimalist Design

TFTP is an ultra-lightweight, UDP-based file transfer protocol engineered to fit into low-memory bootstrap ROMs and BIOS chips.

The Stop-and-Wait Transfer Engine

Operating over **UDP Port 69**, TFTP divides data into fixed 512-byte blocks. Each transmitted block demands explicit acknowledgment before the succeeding block can be released. A block < 512 bytes signals End of File (EOF).

2. Master 3-Way Comparative Matrix

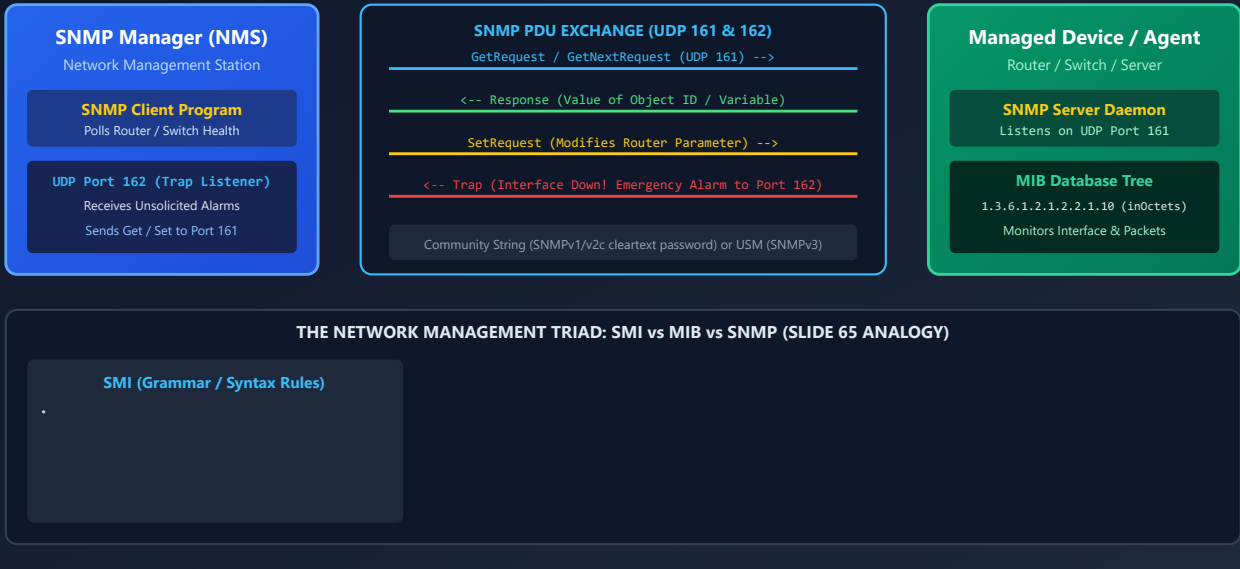
Property	FTP (RFC 959)	TFTP (RFC 1350)	HTTP (RFC 2616)
Transport Protocol	TCP (Port 20 Data, Port 21 Control)	UDP (Port 69)	TCP (Port 80 / 443)
Authentication	Supported (USER / PASS)	None (Zero credentials)	Headers, Basic, Bearer, OAuth
Directory Navigation	Supported (LIST, PWD, CWD)	Not Supported	URL path routing
Application Niche	Enterprise file management	Router firmware loading, PXE boot	Web content delivery & REST APIs

Module 09: Network Management and SNMP Architecture

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

SIMPLE NETWORK MANAGEMENT PROTOCOL (SNMP) FRAMEWORK

Manager-Agent Paradigm, SMI Syntax, MIB Object Hierarchy, UDP Ports 161/162 & PDUs



Analogy: Syntax of C / Java language. • Defines naming conventions using ASN.1. • Defines basic data types (INTEGER, STRING) • Encodes packets via BER (Basic Encoding Rules). MIB (Variable Database) • **Analogy:** Declared variables in program. • Inverted tree structure of managed objects. • Every parameter has a unique Object ID (OID). • e.g., sysUpTime, ifInOctets, ipInReceives. SNMP (Communication Protocol) • **Analogy:** Function calls & message passing. • Defines packet structure & PDU headers. • **3 Core Tasks:** Querying variables (Get), Modifying state (Set), Asynchronous alarms (Trap).

1. SNMP Operational Architecture

Simple Network Management Protocol defines a framework for managing heterogenous devices in an internetwork across a manager-agent architecture.

Transport Layer Port Allocation

UDP Port 161: Managed Agent port accepting polling queries (GetRequest, SetRequest).

UDP Port 162: Manager NMS station port listening for unsolicited asynchronous alarms (Traps).

2. Network Management Triad (SMI vs MIB vs SNMP)

Component	Scope & Semantics	Structural Output
SMI	Grammatical specification and typing syntax	ASN.1 data type definitions (INTEGER, OCTET STRING, OID)
MIB	Structured database of all monitored system variables	Inverted tree of OIDs (e.g., 1.3.6.1.2.1.1 system group)
SNMP	PDU formatting and message passing protocol	Packet frames exchanged over UDP sockets

3. The Three Primary Management Functions

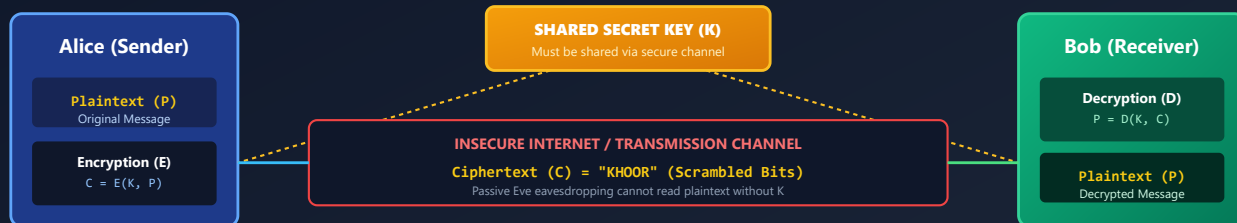
- **Status Polling (GetRequest / GetNextRequest):** Periodic inspection of interface traffic and CPU load.
- **Remote Configuration (SetRequest):** Modifying routing parameters and port operational states.
- **Event Notification (Trap):** Unsolicited emergency alarm reporting critical hardware failures.

Module 10: Cryptography Foundations and Symmetric Ciphers

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

CRYPTOGRAPHY FOUNDATIONS & SYMMETRIC CIPHERS

Confidentiality, Substitution vs Transposition Ciphers, DES & AES Block Cipher Architecture



CLASSICAL vs MODERN SYMMETRIC CRYPTOGRAPHY (SLIDES 73 - 76)

1. Substitution Ciphers

- Characters replaced by other symbols.

Monoalphabetic: Fixed 1-to-1 mapping. Caesar (Shift 3): HELLO -> KHOOR • **Polyalphabetic:** Character mapping changes based on position (Vigenère cipher). • Vulnerable to letter frequency analysis! 2. Transposition Ciphers • Characters retain identity; positions shuffle. • **Rail Fence Cipher:** Plaintext written in zig-zag diagonal rows and read row-by-row. • **Columnar Transposition:** Written in matrix grid and read out based on key order. • Frequency analysis shows standard English letter ratios. 3. Modern Block Ciphers • **DES (Data Encryption Standard):** 64-bit block, 56-bit key, 16 Feistel rounds. (Broken via brute force due to small key). • **AES (Advanced Encryption Standard):** 128-bit block, 128/192/256-bit key. SPN rounds (SubBytes, ShiftRows, MixColumns). • Global standard for military & TLS data.

1. Cryptographic Principles & Security Dimensions

Cryptography ensures secure communication across four pillars: Confidentiality, Integrity, Authentication, and Non-repudiation.

Symmetric Cryptography Architecture

A single secret key K performs both encryption and decryption: $C = E(K, P)$ and $P = D(K, C)$. Requires pre-shared key establishment.

2. Classical Ciphers: Substitution vs Transposition

Cipher Family	Structural Mechanism	Vulnerability Profile
Substitution (Monoalphabetic)	Replaces plaintext letters with static fixed alphabet mapping (e.g., Caesar shift)	Vulnerable to natural letter frequency analysis (e.g., English 'E', 'T')
Transposition (Permutation)	Shuffles letter positions without modifying underlying alphabet characters (Rail Fence, Columnar)	Preserves original character frequency histograms

3. Modern Block Ciphers: DES vs AES

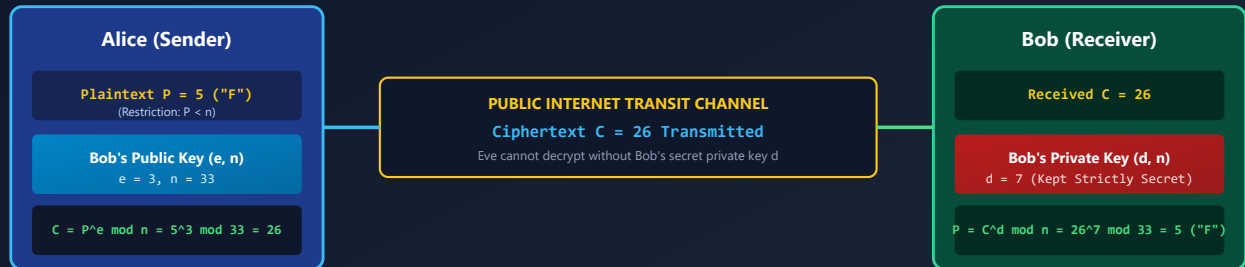
Parameter	DES	AES
Block Size	64 bits	128 bits
Effective Key Length	56 bits (Insecure against modern GPUs)	128, 192, 256 bits (Cryptographically unbreakable)
Round Design	16-Round Feistel Network	10 / 12 / 14 Rounds SPN (SubBytes, ShiftRows, MixColumns)

Module 11: Asymmetric Cryptography and RSA Algorithm

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

ASYMMETRIC CRYPTOGRAPHY & RSA ALGORITHM

Public-Key Mechanics, Prime Modulus Math, Key Generation Steps & Solved AKTU Example



RSA ALGORITHM STEP-BY-STEP MATHEMATICAL FOUNDATIONS (SLIDES 78 - 82)

1. Key Generation Steps

- 1. Choose 2 large primes:

p and **q**. • 2. Compute modulus: $n = p \times q$. • 3. Compute Euler phi: $\phi(n) = (p-1)(q-1)$. • 4. Choose **e**: $1 < e < \phi(n)$ & $\gcd(e, \phi(n)) = 1$. • 5. Compute **d**: $d \times e \equiv 1 \bmod \phi(n)$. 2. Encryption & Decryption • **Public Key**: (e, n) | **Private Key**: (d, n) • **Encryption**: $C = P^e \bmod n$ • **Decryption**: $P = C^d \bmod n$ • **Security Foundation**: Integer Factorization Factoring large n (~2048 bits) is intractable. 3. Solved Numerical Example • $p = 3, q = 11 \rightarrow n = 33, \phi(n) = 20$ • Choose $e = 3$ ($\gcd(3, 20) = 1$) • $d: 3d \equiv 1 \bmod 20 \rightarrow 3 \times 7 = 21 \equiv 1 \rightarrow d = 7$ • Encrypt $P=5$: $C = 5^3 \bmod 33 = 125 \bmod 33 = 26$ • Decrypt $C=26$: $P = 26^7 \bmod 33 = 5$ ("F")

1. Asymmetric Public-Key Cryptography

Asymmetric cryptography eliminates key exchange vulnerability by deploying dual mathematically coupled keys: a universally published Public Key and a confidential Private Key.

The Fundamental RSA Mathematical Engine

Security rests upon the computational difficulty of factoring the product $n = p \times q$ of two massive prime integers.

2. RSA Key Selection Sequence

1. Select distinct large primes p and q .
2. Calculate modulus: $n = p \times q$.
3. Compute totient: $\phi(n) = (p - 1)(q - 1)$.
4. Choose public exponent e such that $\gcd(e, \phi(n)) = 1$.
5. Compute private exponent $d \equiv e^{-1} \pmod{\phi(n)}$.

Encryption: $C = P^e \bmod n$ | Decryption: $P = C^d \bmod n$ ($P < n$)

3. AKTU Solved Numerical: Transmission of "F"

Parameter	Value	Calculation Rule
Prime Inputs	$p = 3, q = 11$	Given prime parameters
Modulus & Totient	$n = 33, \phi(n) = 20$	$n = 3 \times 11, \phi(n) = 2 \times 10 = 20$
Public & Private Exponents	$e = 3, d = 7$	$\gcd(3, 20) = 1, 3 \times 7 = 21 \equiv 1 \pmod{20}$

Plaintext "F"	$P = 5$	0-indexed alphabet code (A=0, ..., F=5)
Ciphertext Result	$C = 26$	$5^3 \bmod 33 = 125 \bmod 33 = 26$
Decrypted Plaintext	$P = 5$ ("F")	$26^7 \bmod 33 = 5$

Module 12: Data Compression Techniques (Lossless & Lossy)

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

DATA COMPRESSION: LOSSLESS vs LOSSY ARCHITECTURE

Run-Length Encoding, Huffman, Lempel-Ziv (LZ), Audio Psychoacoustics & Temporal Masking

LOSSLESS COMPRESSION (Zero Data Loss)

Principle: Exact bit-for-bit mathematical restoration. • **Mandatory For:** Text code (.c, .java), Databases, Executable binaries. • **1. Run-Length Encoding (RLE):** Replaces repeated streaks. "AAAAABBBCCDA" -> "A5B3C2D1A1" (12 bytes -> 10 bytes) • **2. Huffman Coding:** Variable length prefix codes based on freq. • **3. Lempel-Ziv (LZ77 / LZW):** Dictionary table index replaces words. Formats: ZIP, GZIP, PNG, 7z, FLAC, TIFF **LOSSY COMPRESSION (Perceptual Pruning)** • **Principle:** Discards imperceptible details for massive size cuts. • **Mandatory For:** Streaming Video (YouTube), Music, Photos. • **Audio Psychoacoustics & Masking (Slide 87 - 88):** • **Frequency Masking:** Loud tone drowns weaker tone at nearby Hz. • **Temporal Masking:** Loud strike (hammer) desensitizes ear briefly. • **Image Compression (JPEG):** Discrete Cosine Transform (DCT). Formats: MP3, AAC, JPEG, MPEG, H.264, MP4 **DATA COMPRESSION COMPARATIVE MATRIX & METRICS (SLIDE 84 - 88)** 1. Metrics & Equations • **Compression Ratio:** CR = Uncompressed Size / Compressed Size • **Space Savings:** Savings = 1 - (Compressed / Original) 2. Fidelity vs Compression Ratio • **Lossless Ratio:** Modest (usually 2:1 to 4:1). • **Lossy Ratio:** Extreme (10:1 to 50:1+). • Decompressing lossy video does not match original raw RGB sensor matrix bit-for-bit. 3. Psychoacoustic Masking • **Slide 88 Hammer Analogy:** When a loud hammer strikes, ear cannot detect a pin dropping immediately after. • Discarding masked sounds saves 90% space!

1. Data Compression Paradigms

Compression maximizes network transmission throughput and minimizes mass storage consumption by eliminating data redundancy.

Lossless vs Lossy Segmentation

Lossless: Strict bit-for-bit mathematical identity restoration (RLE, Huffman, Lempel-Ziv).

Lossy: Psychoacoustic and visual human perception threshold pruning (MP3, JPEG, H.264).

2. Lossless Compression Schemes

Algorithm	Operational Method	Target Data Stream
Run-Length Encoding (RLE)	Substitutes contiguous identical byte runs with character-count pairs	Scanned monochrome documents, fax bitmaps
Huffman Coding	Builds minimum redundancy binary prefix trees based on symbol probabilities	Variable-length character streams
Lempel-Ziv (LZW)	Maintains dynamic dictionary table substituting recurrent words with pointers	General archive utilities (ZIP, GZIP, PNG)

3. Psychoacoustic Audio Masking Mechanics

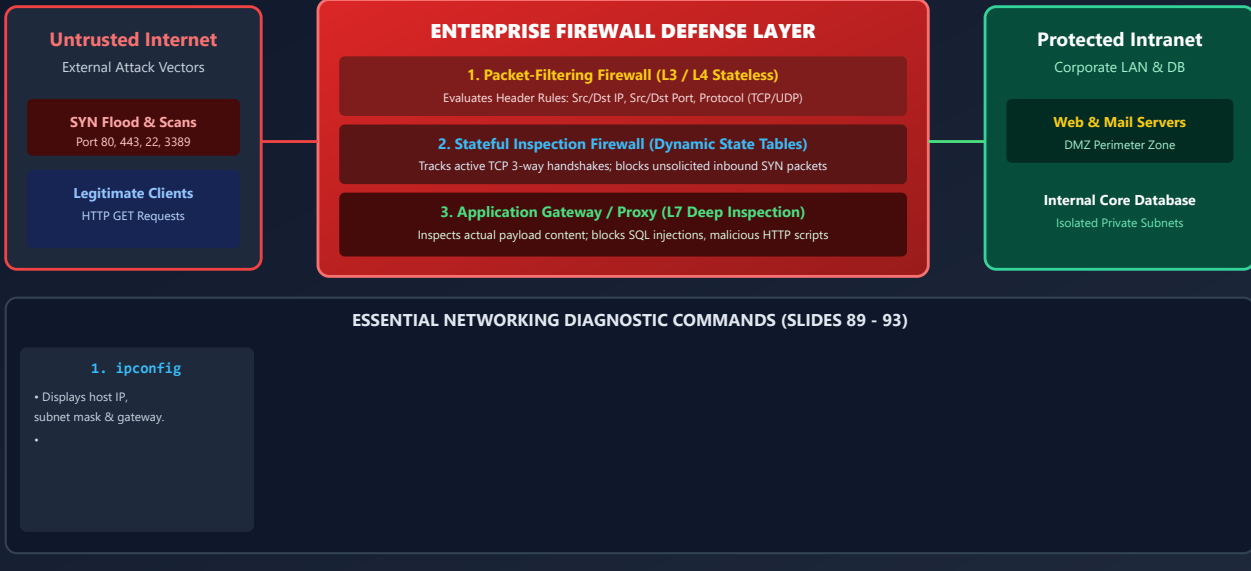
- **Frequency Masking:** A dominant louder sound tone suppresses softer tones residing within the critical spectral band.
- **Temporal Masking (Slide 88):** Loud auditory impulses render the human cochlea unresponsive for several milliseconds immediately preceding and following the transient event.

Module 13: Firewalls, Network Security & Diagnostic Commands

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

FIREWALLS, NETWORK SECURITY & DIAGNOSTIC COMMANDS

Packet Filtering, Stateful Inspection, Application Gateways & Essential Utilities (ping, tracer, netstat)



/all shows MAC & DNS. • /flushdns clears cache. Linux: ifconfig / ip a 2. nslookup • "Name System Lookup". • Queries DNS servers for domain mappings. • Checks MX & A records. e.g., nslookup aktu.ac.in 3. tracer • "Trace Route" utility. • Increments IP TTL (1,2,3) to list intermediate hops. • Identifies network delays. Linux: traceroute 4. ping • Tests end-to-end link. • Sends ICMP Echo Request (Type 8) & waits Reply (0). • Measures packet loss & RTT. e.g., ping 8.8.8.8 5. netstat • "Network Statistics". • Displays active TCP links and listening ports. • -ano shows PIDs & IPs. Essential for port audits.

1. Firewall Defensive Taxonomy

A firewall secures network perimeters by analyzing inbound and outbound packet flows against institutional access policies.

Three Architectural Tiers

- 1. Stateless Packet Filters:** Fast, rule-based matching on L3/L4 headers.
- 2. Stateful Inspection:** Dynamically tracks TCP handshake states to prevent unauthorized external ingress.
- 3. Application Gateways:** Terminates connections and validates application payloads at Layer 7.

2. Five Core Network Diagnostic Utilities (Slides 89 - 93)

Utility	Primary Target	Key Command Syntax
ipconfig	Local adapter IP, subnet mask, default gateway, and DNS configuration	ipconfig /all , ipconfig /flushdns
nslookup	Interactive DNS resolution and zone record inspection	nslookup aktu.ac.in
tracert	Hop-by-hop router discovery and latency profiling using incremental IP TTL	tracert 8.8.8.8
ping	Host reachability and packet loss verification using ICMP Echo Request/Reply	ping aktu.ac.in
netstat	Audit of established TCP socket connections, open listening ports, and PIDs	netstat -ano

Module 14: Unit 5 AKTU Exam PYQs & Numerical Bank

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

UNIT 5: AKTU SEMESTER EXAM PYQS & NUMERICAL BANK

Gateway Classes Past 8 Years Solved Trends (2016-17 to 2023-24) | High-Yield Answer Blueprints

Top High-Frequency AKTU Theory Questions

- 1. TELNET Working & Procedure (4 Times Repeated!)**
AKTU 2016-17, 2017-18, 2018-19, 2023-24: Port 23, NVT character set, IAC.
- 2. DNS Divisions & Resolution (AKTU 2018-19, 2021-22)**
Generic, Country, Inverse domains + Recursive vs Iterative resolution.
- 3. HTTP vs HTTPS & Methods (AKTU 2021-22, 2022-23)**
Port 80 vs 443, SSL/TLS security layer, GET vs POST vs HEAD.
- 4. FTP vs TFTP vs HTTP Comparison (AKTU 2023-24)**
TCP Ports 20/21 vs UDP Port 69 vs TCP Port 80, active vs passive mode.
- 5. SNMP Framework (AKTU 2018-19, 2022-23, 2023-24)**
SMI, MIB, SNMP roles, UDP 161/162, 3 management functions.

Solved Numerical & Protocol Tracing Models

- 1. RSA Character "F" Transmission (AKTU 2022-23)**
 $p=3, q=11, n=33, \phi(n)=20, e=3, d=7, P=5$
Encrypt: $C = 5^3 \bmod 33 = 26$ | Decrypt: $P = 26^7 \bmod 33 = 5$ ("F")
- 2. Caesar Cipher "HELLO" Shift (Slide 75)**
 $C = (P + 3) \bmod 26 \rightarrow H \rightarrow K, E \rightarrow H, L \rightarrow O, L \rightarrow O, O \rightarrow R \rightarrow "KHOOR"$
- 3. Run-Length Encoding Compression Ratio**
Raw: 12 Bytes $\rightarrow$ Compressed: 10 Bytes (CR = 1.2 : 1)
- 4. Passive FTP Port Math**
227 Entering Passive Mode (192,168,1,10, 8, 4) $\rightarrow$ Port = $8 \times 256 + 4 = 2052$

AKTU 10-MARK ANSWER WRITING BLUEPRINT: APPLICATION LAYER

1. Protocol Diagram Mandate

- Always draw client-server interaction diagrams.
- Clearly mark Port numbers: DNS (53), HTTP (80), TELNET (23), SMTP (25), POP3 (110).
- State whether protocol is PUSH or PULL.

2. Mathematical Derivations

- In RSA, explicitly state Euler totient formula $\phi(n) = (p-1)(q-1)$ before computing d .
- Write Extended Euclidean tabular steps.
- Verify decryption step explicitly: $C^d \bmod n$.

3. Comparative Questions

- Use tabular formats for comparison questions:
 - HTTP vs HTTPS | POP3 vs IMAP4
 - FTP vs TFTP | Symmetric vs Asymmetric
- Mention exact RFC standard numbers.

1. High-Frequency Exam Blueprint (AKTU 2016-17 to 2023-24)

Comprehensive solved question repository mapping all repetitive theoretical inquiries and quantitative problems from past university semester examinations.

The Top 5 High-Yield Questions

- TELNET Architecture & NVT Working Procedure (Repeated 4 times: 2016, 2017, 2018, 2023).
- DNS Hierarchy, Divisions & Recursive vs Iterative Resolution.
- Master Comparisons: HTTP vs HTTPS, FTP vs TFTP, POP3 vs IMAP4.
- SNMP Triad: SMI, MIB, SNMP & Three Management Functions.
- RSA Key Generation Derivation and Numerical Computation (Character 'F').

2. Solved Numerical Problem Review

Problem Type	Input Parameters	Mathematical Output
RSA Character "F" Transmission	$p=3, q=11, e=3, P=5$	$n=33, \phi(n)=20, d=7, C=26$, Decrypted $P=5$ ("F")
Passive FTP Data Port Extraction	227 Passive (192,168,1,50, 16, 32)	Port = $16 \times 256 + 32 = 4128$
Caesar Cipher Monoalphabetic Shift	Plaintext = "HELLO", Shift $k = 3$	Ciphertext = "KHOOR"